

Feb 1, 2024.

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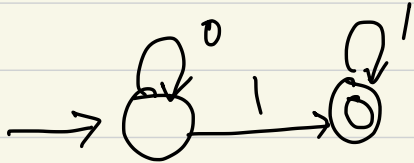
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W3L2

$$B = \{0^n 1^n \mid n \geq 0\} \quad \Sigma = \{0, 1\}$$

$$L(B) = \{\epsilon, 01, 0011, 000111, \dots\}$$

Can we construct a DFA/NFA for B



```
bool B(string w) {  
    int L = length(w);  
    if (L % 2 != 0) return false;  
    else {  
        // check first half for 0's  
        // check 2nd half for 1's  
        // if all is well  
        // return true  
        // else return false  
    }  
}
```

$C = \{w \mid w \text{ has an equal no. of } 0\text{'s} \text{ \& } 1\text{'s}\}$   
 $D = \{w \mid w \text{ has an equal no. of } 01 \text{ \& } 10 \text{ as substrings}\}$

|    | C            | D   |         |
|----|--------------|-----|---------|
| 1) | yes          | yes | 2       |
| 2) | yes          | no  | same    |
| 3) | no           | yes | fewer   |
| 4) | no           | no  | most    |
| 5) | I'm confused |     | several |

$D = \{w \mid w \text{ has an equal no. of } 01 \text{ \& } 10 \text{ as substrings}\}$

$D = \{\epsilon, \underline{010}, 0110, 01110, \underline{0111\ldots10}, \ldots$

~~00~~ 0, 00, 000, . . .

1, 11, 111, . . .

101, 1001, . . . }

$A \rightarrow B$  $B \rightarrow A$ yes/no $A \rightarrow B$  $\neg A \rightarrow \neg B$ 

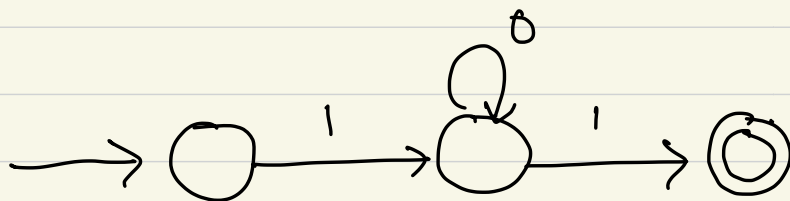
yes/no?

 $A \rightarrow B$  $\neg B \rightarrow \neg A$ yes/no

Pumping Lemma: for regular languages

If  $A$  is a RL, then there is a number  $p$  (pumping length) where if  $s$  is any string in  $A$  of length at least  $p$ , then  $s$  may be split/divided into three pieces  $s = xyz$ , satisfying following conditions:

- 1) for each  $i \geq 0$ ,  $xy^iz \in A$ ,
- 2)  $|y| > 0$ , ad
- 3)  $|xy| \leq p$ .

 $N$  $p=3$  $s = 101 = xyz$  $x=1, \underline{y=0}, z=1$  $\rightarrow 1001, 10001, 10...01$  $\rightarrow 11 \in L(N)$

$$\underline{B = \{0^n 1^n \mid n \geq 0\}}$$

Proof that B is not regular.

1. Assume B is regular

$\Rightarrow$  we can PL on B.

assume  $p$  is the pumping length.

$$S = 0^p 1^p \quad |S| > p$$

so PL we can divide  $S$  into three parts/pieces.  $S = xyz$

①  $\forall i \geq 0, xy^i z \in A$

②  $|y| > 0$  ✓

③  $|xy| \leq p$  ✓

$$x = 0^{p-1}, \quad y = 0, \quad z = 1^p$$

①

$$i = 0$$

$$i = 1, 2, 3, \dots$$

$$0^{p-1} 1^p \notin B. ?$$

$$0^{p-1} 00 1^p \in B.$$